

Sparky Voice AI - System Tray Edition

A fully local, open-source voice assistant with wake word detection. No cloud dependencies, no accounts required.

Features

- ✔ **Wake Word Detection** - Always listening for "Hey Jarvis" (or custom "Hey Sparky")
- ✔ **System Tray Integration** - Runs minimized in system tray
- ✔ **Conversation Mode** - Active listening between wake/exit words
- ✔ **Auto-Start** - Optional Windows startup automation
- ✔ **Custom Wake Words** - Train your own wake words locally
- ✔ **100% Local** - No cloud services, no accounts, no expiration

Quick Start

1. Install Dependencies



bash

```
pip install -r requirements.txt
```

2. Configure

Edit config.ini with your server details:



ini

```
[VoiceAI]
server_host = 10.6.1.15
server_port = 8004
default_voice = ara

[LLM]
llm_host = 10.6.1.15
llm_port = 8000
llm_api_key = sparky-secret-key
llm_model = OpenHermes-2.5-Mistral-7B
```

3. Run



bash

```
python voice_client_tray.py
```

The app will appear in your system tray with a green microphone icon.

Usage

System Tray Menu

Right-click the tray icon to access:

- **Status** - Current listening state
- **Start/Stop Listening** - Toggle wake word detection
- **Wake/Exit Words** - View current wake words
- **Train Custom Wake Words** - Instructions for training
- **Reload Wake Words** - Reload models after training
- **Start with Windows** - Toggle auto-start
- **About** - System information
- **Quit** - Exit application

Default Wake Words

By default, the app uses built-in models:

- **Wake word:** "Hey Jarvis"
- **Exit word:** "Hey Mycroft"

Conversation Flow

1. Say **"Hey Jarvis"** to activate
2. Assistant responds: "Yes? How can I help you?"
3. Have your conversation
4. Say **"Hey Mycroft"** to deactivate
5. Assistant responds: "Goodbye!"

Training Custom Wake Words

Option 1: Google Colab (Easiest)

1. Open the [openWakeWord Colab notebook](#)
2. Train two models:
 - Model 1: "hey sparky"
 - Model 2: "bye sparky"
3. Download the .tflite files
4. Rename them:
 - hey_sparky.tflite

- bye_sparky.tflite
- 5. Place in wake_models/ directory
- 6. Right-click tray icon → **Reload Wake Words**

Option 2: Local Training (Advanced)

Coming soon - automated local training script.

Icon Colors

-  **Green** - Listening for wake word
-  **Red** - Active conversation
-  **Blue** - Processing
-  **Orange** - Speaking
-  **Gray** - Idle/stopped

Auto-Start Setup

1. Right-click tray icon
2. Select **Start with Windows**
3. App will launch automatically on boot

To disable, uncheck the same menu item.

Troubleshooting

Wake word not detecting

- Check microphone permissions
- Adjust microphone volume (should be clearly audible)
- Try speaking closer to microphone
- Check audio device in Windows settings

Models not loading



bash

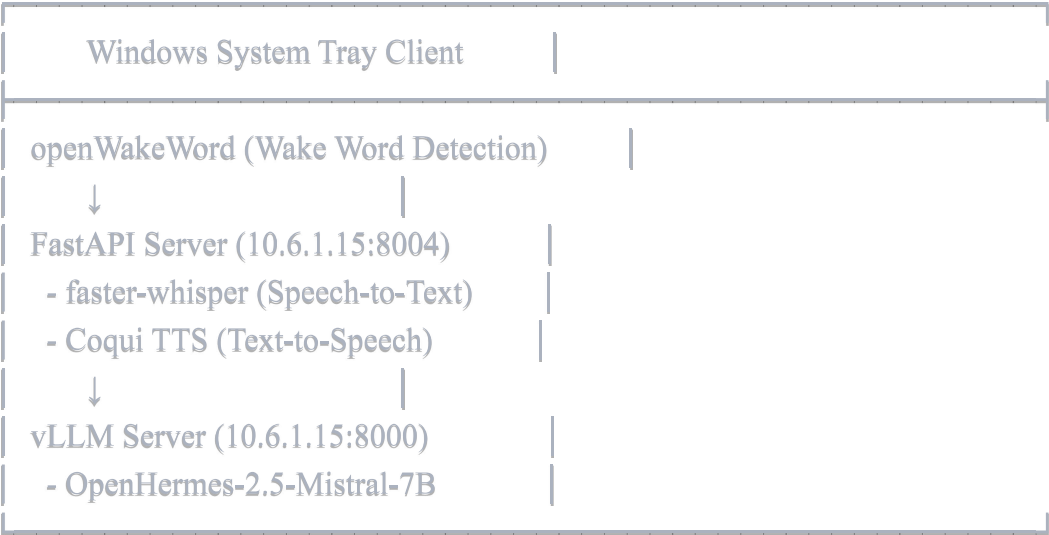
Download built-in models manually

```
python -c "import openwakeword; openwakeword.utils.download_models()"
```

Custom models not working

- Ensure .tflite files are in wake_models/ directory
- File names must be exact: hey_sparky.tflite and bye_sparky.tflite
- Use **Reload Wake Words** from menu after adding files

Architecture



File Structure



```
sparky-voice-ai/
├── voice_client_tray.py  # Main tray application
├── config.ini           # Configuration
├── requirements.txt     # Python dependencies
├── wake_models/        # Custom wake word models
│   ├── hey_sparky.tflite # (optional, user-trained)
│   └── bye_sparky.tflite # (optional, user-trained)
└── README.md           # This file
```

Dependencies

- **openwakeword** - Wake word detection
- **sounddevice** - Audio capture
- **soundfile** - Audio file handling
- **requests** - HTTP client
- **pystray** - System tray integration
- **Pillow** - Icon generation
- **pywin32** - Windows integration

Future Enhancements

- ☐ Automated local wake word training
- ☐ Voice Activity Detection (VAD) for better command capture
- ☐ Multiple voice profiles
- ☐ Conversation history persistence
- ☐ Custom response sounds
- ☐ Performance metrics dashboard
- ☐ Linux/Mac support

License

Open-source - use and modify as needed for personal projects.

Support

For issues or questions, check:

- openWakeWord: <https://github.com/dscripka/openWakeWord>
- Your FastAPI server logs
- Windows Event Viewer

Built with  for local-only, privacy-focused voice AI